Data Preparation

Date	28 october 2026
Team ID	LTVIP2026TMID00420
Project Name	EV Market Analysis Using Tableau
Maximum Marks	5 Marks

Data preparation refers to the process of cleaning, standardizing, transforming, and organizing data so that it is suitable for analysis and reporting. In a Snowflake project, data preparation involves working with the ingested RAW data, identifying data quality issues, creating STAGING tables, applying required transformations, and preparing reliable data for the CURATED and MART layers.

Activity 1:

Null ValuesEV dataset me Charging_Time aur Customer_Rating columns me kuch null values thi. Tableau me Data Interpreter on karke aur Excel me Filter laga ke blank rows ko remove / average se fill kiya.

Code: IFNULL([Battery_Range], AVG([Battery_Range]))

Output: 0 Null values remaining

Activity 2:

Duplicate HandlingModel_ID column ke basis par duplicate check kiya. 12 duplicate rows mili jo same car model ki repeat entry thi.

Code: Tableau me Data -> Remove Duplicates

Output: Dataset reduced from 15890 to 15878 rows.

Activity 3:

Cleaning & StandardizationManufacturer column me TATA, Tata Motors, tata alag-alag likha tha. Isko standard Tata Motors kiya. Price column me $ aur , hata ke number banaya.

Output: All Manufacturer names standardized.

Activity 4:

Transformation & ValidationNew calculated fields banaye: Price_Category, Sales_Year = YEAR([Purchase_Date]), Range_Category.

Output: Dataset validated and ready for 9 visualizations.